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Neuromuscular differences between boys with and without intellectual disability during squat jump

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ABSTRACT

The purpose of this study was to identify the differences in vertical squat jump (SJ) between volunteers with and without intellectual disability (ID). Thirteen boys with ID (average intelligence quotient, estimated by Wisk III test: 55.6 ± 11.2) and 13 peers without disabilities performed maximal SJ on a force platform. Kinematic data were captured using a six-camera 3D motion analysis system and electromyographic (EMG) activity was recorded using surface electrodes. Unpaired T-test determined the statistical difference between the two groups. The obtained results indicated that the group with ID, jumped lower, developed lower vertical ground reaction forces, knee power output, knee angular velocity, and take-off velocity, and showed longer propulsion duration, decreased mean to maximum agonist EMG activity and higher antagonist/agonist activity ratio. The deficit in the SJ observed in individuals with ID was attributed to a deficit in the examined mechanical and neuromuscular parameters, and especially to the agonist and antagonist co-contraction.

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1. Introduction

Intellectual disability (ID) is one of the most prevailing reasons of disability in children. Its case and neurophysiological development are still poorly understood (Yu et al., 2008). Among the deficits on various functional domains such as cognition, language, audition and moral judgment, ID has also negative effects in gross and fine motor performance, in terms of motor reaction time (LeClair, Pollock, & Elliott, 1993) and balance (Vuijk, Hartman, Scherder, & Visscher, 2010) or when performing a visuo-manual rhythmic tracking task (Van Aken et al., 2010). A possible neurophysiological mechanism leading to such motor deficits in people with ID could be the damage in the integrity of the cortico-spinal tract and the changes in morphometry of the frontal and paretial cortex in people with lower intelligence quotient (Haier, Jung, Yeo, Head, & Alkire, 2004; Yu et al., 2008). It is important to note though, that the motor deficit of people with ID could be influenced and deteriorated by behavioral factors, such as sedentary life (Horvat & Franklin, 2001) which might have several consequences. Previous studies have shown that individuals with ID, exert less strength and power (Angelopoulou, Tsimaras, Christoulas, Kokaridas, & Mandroukas, 1999; Horvat, Pitetti, & Croce, 1997; Horvat, Croce, Pitetti, & Fernhall, 1999), and have poorer skill performance (Horvat, Croce, Zagrodnik, Brooks, & Carter, 2012; Skowronski, Horvat, Nocera, & Croce, 2009) compared to peers without ID.

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An earlier study highlighted the importance of physical fitness to promote health and well-being of persons with ID (Fernhall, Tymeson, & Webster, 1988). Recently, Hinckson, Dickinson, Water, and Sands (2013) showed that children with ID can benefit from training programs in order to manage their weight. This finding in conjunction with the fact that children with ID have an increased risk for obesity (Maiano, 2010) support the importance of exercise, especially for these children.

During the past decades team sports are becoming more popular for people with ID. Such activities consist a very useful medium to increase physical activity and develop or enhance various motor skills, which might have positive impact on the activities of daily living (Watkinson et al., 2001). Basketball is a popular team sport which incorporates various motor skills such as running, dribbling, shooting, ball passing/receiving and jumping. This implies that targeted training programs improving these skills have to be developed. In order to achieve this, it is important to understand the causes of possible motor deficits on individual motor tasks, by describing in detail their properties.

The investigation of jumping in people with ID is very limited, although it is a common task used in their training, competition and daily life. It has been previously reported that people with ID are inferior in standing long jump distance (Skowronski et al., 2009). Jump is a very complex task and depends on many factors, such as the coordination and energy flow through the joints of the lower extremities (Bobbert & van Ingen Schenau, 1988; Pandy & Zajac, 1991; Prilutsky & Zatsiorsky, 1994), as well as the morphology of the muscle tendon unit (Bobbert, 2001; Bosco, Tihanyi, Komi, Fekete, & Apor, 1982), the muscle activation patterns (Bobbert & van Soest, 2001; Bobbert & van Zandwijk, 1999), and the mechanical power output (Canavan & Vescovi, 2004; Kollias, Panoutsakopoulos, & Papaiakovou, 2004; Vanezis & Lees, 2005). Studies on jumping performance of persons with ID seem to be limited. However, taking into account that people with ID have similar adaptation responses after training programs (Carmeli, Zinger-Vaknin, Morad, & Merrick, 2005; Hemayattalab & Movahedi, 2010), it is essential to understand the reasons that cause any performance deficit in individuals with ID, by describing their neuromuscular properties during jumping and comparing them to peers with no ID.

Therefore, the purpose of this study was to examine the neuromuscular differences in vertical squat jump (SJ) between boys with and without ID. The results will provide evidence for the causes of the performance deficit in boys with ID and will help in the future to develop strategies in order to improve their performance by designing more targeted and specific training programs. Such training interventions are becoming more necessary if we consider the increasing number of people with ID who participate systematically in competitive sports that have jumping as a crucial task determining the athlete's performance.

2. Methods

2.1. Participants

Twenty-six boys (14–17 years old) participated voluntarily in the study. The participants with ID ($n = 13$, mean age: 15.3 ± 1.6 years, body mass: 56.3 ± 15.7 kg, body height: 164.2 ± 11.3 cm) had an intelligence quotient of 55.6 ± 11.2, as estimated by Wisk III test. The group without ID ($n = 13$) had mean age 15.4 ± 1.2 years, body mass 58.5 ± 12.5 kg, and body height 168.5 ± 11.5 cm.

A medical board consisted of a physician, a psychiatrist and a neurologist, verified that the participants were free of any neurological motor deficit or any injury history on the back or lower extremities. All participants agreed to participate in the study after being informed about the procedures. Before testing, they and their parents read and signed a written informed consent statement. The participants were asked to abstain from any exercise activity for 48 h before the examination. The experimental procedures were according to the ethic guidelines of the Aristotle University of Thessaloniki, Greece.

2.2. Experimental procedure

All participants were informed verbally about the procedure. Two weeks before the main protocol, they visited the laboratory for familiarization, including execution of SJ from a starting position of 90° knee flexion, as previously suggested (Bobbert, Casius, Sijpkens, & Jaspers, 2008). When they visited the laboratory, after orientation in space, the evaluation of anthropometric parameters was assessed. Afterwards they performed a general warm up (5 min pedaling with selected pace on a bicycle and warming up exercises) followed by a specific warm up, including submaximal vertical jumps with increasing intensity. Before the final assessment three maximal SJ were performed for further familiarization and three extra jumps were performed for the evaluation. The best jump in height was selected for further evaluation. To avoid any fatigue effect, the interval between SJ was set at 3 min.

2.3. Instruments and testing

Kinematic data were captured at 100 Hz using a six-camera 3D motion analysis system (VICON 612, Oxford Metrics Ltd., Oxford, Oxfordshire, UK). Vertical Ground Reaction Forces (vGRF) were recorded with a ground-mounted 40 cm × 60 cm force plate (Bertec Type 4060, Bertec Corporation, Columbus, OH, USA). The electromyogram (EMG) was recorded using a BTS Telemg device (Milano, Italy).

In the testing day after assessment of the physical characteristics the EMG electrodes were located on the dominant limb, as determined by the preference limb to kick a ball (Wikstrom, Tillman, Kline, & Borsa, 2006). The EMG electrodes were

placed according to the SENIAM instructions (Hermens, Freriks, Disselhorst-Klug, & Rau, 2000). The vastus lateralis (VL) and the biceps femoris (BF) EMG activity were recorded, responding to the agonist and antagonist muscle activity, respectively. The ground electrode was placed over the bony surface of the contra lateral wrist. Skin was initially shaved and cleaned with an alcohol solution, achieving skin impedance below 2 kΩ. EMG signals were recorded with 2 kHz sampling rate, filtered with zero lag, second order, band-pass (10–500 Hz) Butterworth filter, followed by full-wave rectification and then low-pass filtering at 50 Hz.

For the kinematic analysis 16 reflective spherical markers (14 mm in diameter) were placed on anatomical bony landmarks of the lower extremities according to Davis, Õunpuu, Tyburski, and Gage (1991). Prior to data collection session, the motion analysis system was calibrated according to manufacturer's recommendations. Trajectories were filtered with the generalized cross validated splines (Woltring & Fortran, 1986).

SJ was performed on the force plate. The participants were standing and slowly bent their knees until they reached a knee flexion of 90°. They remained at this position for 2 s to eliminate any counter movement effect. Before the trials participants were instructed to "jump as high as possible".

2.4. Data analysis

Recording data were further analyzed using Matlab 6.1 scripts (The MathWorks Inc.). The peak vGRF and the power of knee joint during propulsion (P_{knee}) were normalized to the body weight of each individual. Moreover, the mean angular velocity of the knee joint during the propulsion phase (AV_{knee}) and the take-off velocity (V_{to}) were estimated. The average EMG amplitude of VL was calculated over the propulsion phase and was normalized to the maximum EMG value (Yang & Winter, 1984). The co-activation of the BF was expressed as the ratio of BF to VL muscle activity (Lazaridis et al., 2010).

2.5. Statistical analysis

Average values and standard deviation of all dependent variables were calculated. Unpaired T-test between the two groups was applied for all dependent variables. Statistical analysis was performed with the SPSS 16.0 (SPSS Inc.). The level of significance was set at 0.05.

3. Results

Raw data from one typical example from each group is presented in Fig. 1.

The jump height of the group with ID was lower compared to the group without disabilities (Fig. 2). This difference was statistically significant ($t_{24} = 3.02$, $p < 0.05$). Similarly, the normalized vGRF and P_{knee}, were significantly lower in the group with ID compared to the group without disabilities ($p < 0.05$, Table 1).

Regarding the kinematic data during propulsion phase, the group with ID demonstrated significant lower values in AV_{knee} ($t_{24} = 2.64$, $p < 0.05$), and V_{to} ($t_{24} = 2.99$, $p < 0.05$), compared to the group without disabilities.

Relative to the EMG data during the propulsion phase (Fig. 3), the group with ID had significantly lower EMG activity in both muscles (VL: $t_{24} = 2.51$, $p < 0.05$, and BF: $t_{24} = 3.39$, $p < 0.05$). The level of co-activation was higher in the group with ID compared to the group without ID ($t_{24} = 3.92$, $p < 0.05$, Fig. 4).

4. Discussion

The obtained results indicated that children with ID jumped lower, and developed lower vGRF, P_{knee}, AV_{knee} and V_{to} compared to their counterparts without disabilities. Furthermore, they had longer propulsion phase, decreased agonist EMG activity, and higher antagonist to agonist EMG ratio.

To our knowledge, there are no previous studies examining the SJ in children with ID and analyzing the possible reasons for their lower performance. There is only one relevant study showing that children with ID achieved lower score in standing long jump (Skowronski et al., 2009). Considering that SJ and standing long jump are in principal similar (Ridderikhoff, Batelaan, & Bobbert, 1999), it is reasonable that our results are in agreement with the above-mentioned study. The obtained data could be supported by reports stating that individuals with ID develop lower strength and power (Chia, Lee, & Teo-Koh, 2002; Horvat et al., 1999), and present deficit in coordination during multijoint movements, such as gait (Horvat et al., 2012). Damages in the higher sensori-motor pathways and the cortico-spinal tract shown with neuro-imaging studies in people with ID (Yu et al., 2008) could be a possible explanation for the above-mentioned deficits in motor performance. The present study, present further data that could explain the decreased SJ performance in children with ID.

Regarding the mechanical parameters, the group with ID presented lower values in the vGRF, P_{knee}, AV_{knee} and V_{to}. The importance of these parameters for jumping performance has been documented earlier (Kollias, Hatzitaki, Papaiakovou, & Giatsis, 2001; Vanezis & Lees, 2005). There are no comparable data in the literature for persons with ID. However, the obtained data could be supported by previous studies showing lower power under isokinetic conditions (Horvat et al., 1999) and after performing the Wingate test (Chia et al., 2002) in people with ID. Considering the above it could be argued that the decreased SJ performance in children with ID could be attributed to their reduced ability to produce an adequate amount of force and power during the propulsion phase.

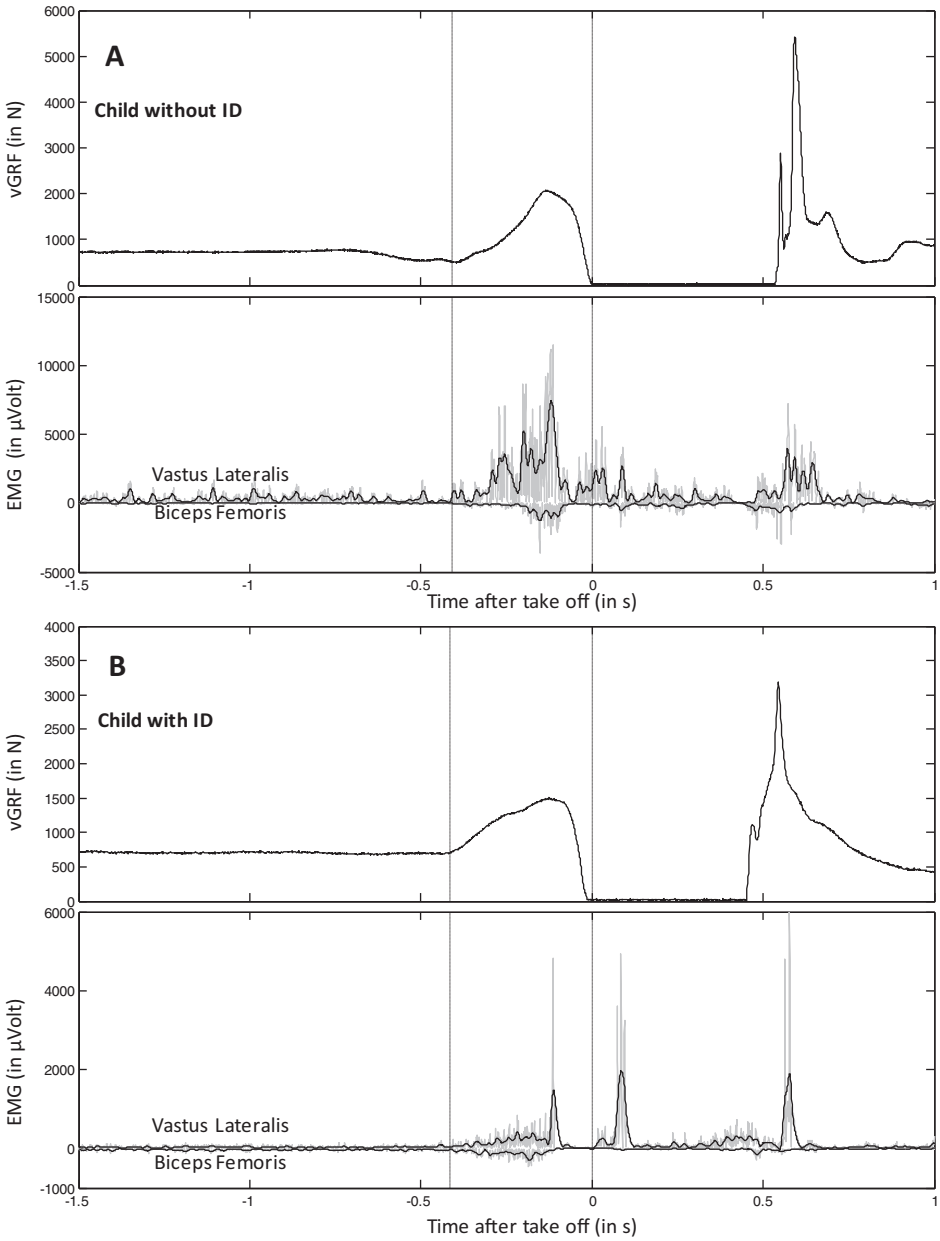

Fig. 1. Raw data of vertical ground reaction force (vGRF) and electromyographic activity (EMG) of the vastus lateralis and biceps femoris (shown inverted) during squat jump of an individual without and with ID (A and B, respectively). Rectified EMG signal is presented in gray. Vertical doted lines define the start and end of the propulsion phase.

The importance of internal and external motivation of persons with ID during tests that aim to maximize motor performance is an important limitation of the current study. In a study examining muscle fatigue during intermittent exercise, people with ID resisted more to fatigue (Zafeiridis et al., 2010) and it was argued that persons with ID might not be able to activate their motor units in the same extend as people without ID. This speculation requires further investigation, using methods such as twitch interpolation technique (Herbert & Gandevia, 1999) and might have implications in the current study as well; although encouragement to maximize performance was common in both examined groups we are still not certain to what extent maximum power output was achieved. Therefore, techniques to control external and internal motivation should be considered in future studies.

The lower efficiency of the group with ID during jumping could be interpreted based on data of experienced and less experienced persons who perform complex movements. According to Bobbert and van Soest (2001), the optimal agonist activity is a crucial factor to maximize jump height performance. Horita, Komi, Nicol, and Kyröläinen (2002) showed that experienced jumpers activate their agonist muscle more than less experienced ones. Furthermore, comparisons between

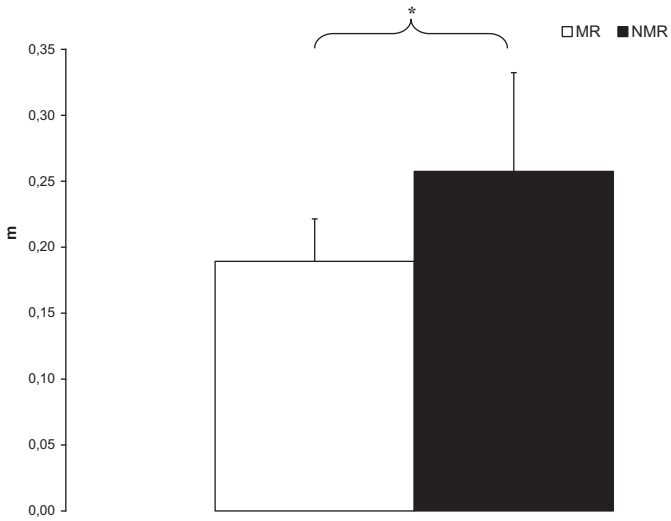

Fig. 2. Mean ± SD of performance in vertical squat jump (jump height in meters) for the group with ID and the group without ID (*$p < 0.05$).

Table 1
Values (mean ± standard deviation) of kinematic parameters during vertical squat jump for the boys with and without ID. [vGRF: peak vertical ground reaction force; P_{knee}: peak knee power; AV_{knee}: mean knee angular velocity during propulsion; V_{to}: take-off velocity].

Group	vGRF (N/kg)	P_{knee} (W/kg)	AV_{knee} (°/s)	V_{to} (m/s)
With ID ($n = 13$)	1.0 ± 0.3	17.7 ± 6.6	108.1 ± 34.9	1.9 ± 0.2
Without ID ($n = 13$)	1.3 ± 0.4*	25.4 ± 9.4*	155.6 ± 52.0*	2.2 ± 0.3*

* $p < 0.05$.

children and adults in various jumping tasks have shown that children activated less their agonist muscle than adults (Lazaridis et al., 2010; Lloyd, Oliver, Hughes, & Williams, 2012). Regarding muscle activation in people with ID, limited studies lead to controversial results. Horvat, Ramsey, Amestoy, and Croce (2003) supported that children with ID present similar EMG amplitude, variability, and median power spectrum frequency as their counterparts without ID when performing simple reaction tasks. On the contrary, earlier studies examining the agonist activity of persons with ID showed

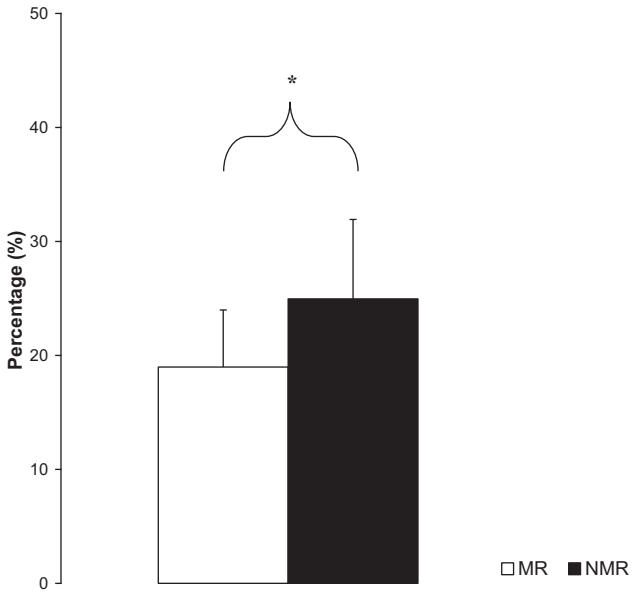

Fig. 3. Mean ± SD of EMG activity (% of the maximum EMG value) of the vastus lateralis (VL) muscle during the vertical squat jump for the prepubescents with (ID) and without (nID) intellectual disability (*$p < 0.05$).

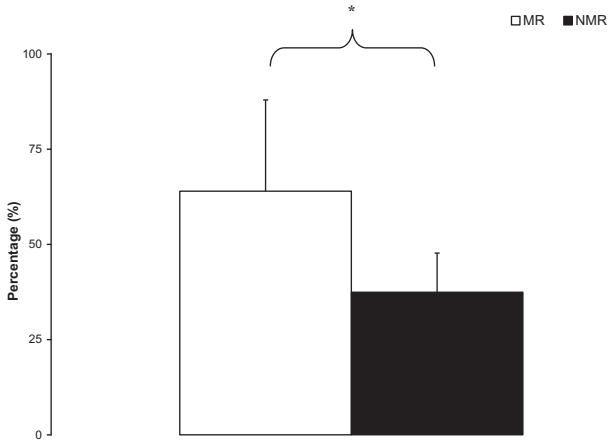

Fig. 4. Mean ± SD of biceps femoris muscle coactivation (expressed as percent of the vastus lateralis EMG) during the propulsion phase of the vertical squat jump for the prepubescents with (ID) and without (nID) intellectual disability (*$p < 0.05$).

that the level of agonist activation is lower and more variable (Davis, Sparrow, & Ward, 1991). Our study is in agreement with the latter one, showing decreased mean agonist activity normalized to maximum EMG during SJ in individuals with ID. Hence, it seems that there is no consensus in the literature regarding agonist activity in ID, but our data regarding SJ support the concept that children with ID produce lower agonist EMG which is a characteristic of inexperience jumpers, which implies that this behavior could be reversible.

Regarding the antagonist co-activation, which was higher in the group with ID, the only available study concerning maximal isokinetic contractions has shown higher antagonist activity in people with ID as well (Tsimaras et al., 2009). Principally, antagonist activity is considered as joint stiffness regulator (Kellis, 1998) and is related to the maturation and the process of learning (Hamstra-Wright et al., 2006). In general, the higher co-activation could be attributed to lower coordination when performing complex movements (Chia et al., 2002), provided that no neurological disability is apparent, as shown in children with cerebral palsy (Unnithan, Dowling, Frost, Ayub, & Bar-Or, 1996). Strong evidence for this learning effect is supplied by studies comparing children and adults, with the former presenting higher co-activation during jumping (Lazaridis et al., 2010) and running (Frost, Dowling, Dyson, & Bar-Or, 1997).

The current data imply that the lower jump height achieved from children with ID, is related to the forces produced mainly from the extensor (antigravity) muscles of the lower limbs, as shown by the reduced vGRF and more particularly, by the decreased P_{knee} and AV_{knee} which is the result of net torque produced by the muscles around the knee joint. Furthermore, the EMG data of the agonist and antagonist muscles illustrated that the observed deficit in the mechanical properties of the knee, could be of neural nature. Keeping in mind that our sample had no neurological symptoms, it could be argued that learning processes could take place. Considering the above and taking into account that the deficit in motor performance for people with ID has been attributed to environmental factors as well (Horvat & Franklin, 2001), it is recommended that training improving coordination and strengthening of the lower limbs could enhance jumping performance in people with ID. More specifically, it is suggested that children with ID, free of any neurological sign could be trained to improve their technique, as well as their strength and power, by performing strength and plyometric training, as recommended in children without any disability (Harries, Lubans, & Callister, 2012; Johnson, Salzberg, & Stevenson, 2011). The application of such protocols is supported by recent studies which have shown that children with ID improved their lean body mass and bone mineral density after a plyometric and combined training protocol (González-Agüero, Vicente-Rodríguez, Gómez-Cabello, Ara, Moreno, & Casajús, 2011; González-Agüero, Vicente-Rodríguez, Gómez-Cabello, Ara, Moreno, & Casajús, 2012).

5. Conclusion

The current study confirms the deficit of people with ID to perform multijoint movements, such as the SJ. The deficit in this task was attributed to mechanical and neuromuscular parameters. It is suggested that the principles of training for people without ID could be applied in people with ID as well, provided they don't suffer from any neurological symptom. However, in order to have a more holistic description of the jumping skill, further research is required in persons with ID, by examining additional parameters such as muscle fiber distribution, stiffness regulation of the muscle-tendon unit and energy flow through the joints.

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